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Compact Motor Envelope and Sizing Worksheet

For machine designers fitting a DC motor and gearhead into a limited space

Complete the steps in order. Steps 1 and 2 define the space and the load, Steps 3 and 4 choose the motor speed, ratio and technology, and Steps 5 and 6 confirm that the result stays cool and really fits.

Step 1 — Define the envelope

- [ ] Maximum length and diameter or square section available for motor, gearhead and connector together
- [ ] Whether the space is length-limited, diameter-limited or both
- [ ] Mounting face location and required mounting type (B5 flange, or B14 face for tight spaces)
- [ ] Output shaft position and axis: in line, offset or at right angles to the motor
- [ ] Keep-out zones, nearby heat sources and parts that move during operation

Step 2 — Define the load at the output

- [ ] Continuous output torque (Nm) and speed (rpm) at the operating point
- [ ] Peak and starting torque, and how long each lasts
- [ ] Output power: $P = T \times N \times 0.1047$ (W, with T in Nm and N in rpm)
- [ ] Duty type: S1 continuous, S2 short-time or S3 intermittent with cyclic duration factor
- [ ] Radial and axial load on the output shaft and their positions

Step 3 — Trade speed for size

- [ ] Choose a motor speed, commonly about 3,000 rpm for small PMDC and BLDC motors
- [ ] Ratio: i = motor speed / output speed, rounded to an available ratio
- [ ] Motor torque: Tm = Tout / (i × gear efficiency)
- [ ] Rotor volume falls roughly by the factor i × gear efficiency compared with direct drive
- [ ] Check gearhead continuous and peak output ratings against the load with margin for starting

Step 4 — Choose gearhead geometry and motor technology

- [ ] Long, narrow space: in-line planetary gearhead, typically around 90–97 % per stage
- [ ] Wide space or wheel drive: parallel-shaft spur or helical, roughly 95–98 % per stage
- [ ] Length blocked by an obstruction: right-angle worm or bevel; worm efficiency roughly 40–90 %
- [ ] Short, wide space: flat, short-stack motor, since torque scales with diameter squared
- [ ] BLDC for highest torque density and long life, if space exists for the driver and wiring
- [ ] Ferrite PMDC for simple DC supply, intermittent duty and lowest system cost

Step 5 — Thermal check

- [ ] Measure the air temperature inside the enclosure at the motor, not the room temperature
- [ ] Compare candidates by motor constant Km = Kt / square root of R; copper loss = $(T / Km)^2$
- [ ] Provide a conduction path: face-mount onto a metal bracket or plate
- [ ] Allow for magnet weakening when hot: ferrite about 0.18–0.2 % per °C, NdFeB about 0.1–0.12 % per °C

- [ ] Confirm with a temperature-rise test at the real duty and ambient

Step 6 — Interfaces and service access

- [ ] Add the length of any brake, encoder and the connector plus cable bend radius
- [ ] Specify lead exit as axial or radial, with a clock angle relative to a mounting hole
- [ ] Leave clearance to remove brush caps on PMDC motors with externally accessible brushes
- [ ] Check shaft extension, key or flat, and coupling or pinion engagement
- [ ] Send a STEP model of the surrounding parts with a dimensioned 2D envelope drawing

Drive options for compact envelopes

Option	Envelope shape	Strength	Watch-out
Motor + planetary	Slim coaxial cylinder	Highest gear torque density	Backlash; cost per stage
Motor + parallel-shaft	Offset output, box shape	Efficient and robust	Wider; offset output axis
Right-angle worm or bevel	L-shape, short axially	Folds around obstructions	Worm efficiency 40–90 %
Flat short-stack motor	Wide and short	Uses diameter, strongest lever	Higher rotor inertia
Direct drive	Large in all directions	No backlash or gear wear	Lowest torque per volume at low speed

Efficiency ranges are typical and vary with ratio, size, lubrication and load. Always confirm continuous torque, gearhead ratings and temperature rise against manufacturer data and a test at the real duty cycle.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.